

Wavelet Analysis of Sunspot Activity: Exploring Rates of Change and Acceleration

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This study presents a detailed time–frequency analysis of daily sunspot numbers spanning 2000–2023, with the aim of characterizing the dynamical behavior of solar activity across multiple temporal scales. Owing to the pronounced nonstationarity and cyclic structure of the raw series, first- and second-order differencing are employed to better isolate underlying temporal dynamics. The resulting first-order (D1) and second-order (D2) series capture the rate of change and the acceleration of sunspot activity, respectively, with stationarity tests confirming their improved suitability for spectral analysis.

Frequency-domain analysis via periodograms indicates that the D1 series retains multiple statistically significant periodic components, with dominant peaks near 2.5, 9.8, and 22.6 days. These periodicities reflect a combination of high-frequency variability and intermediate-scale processes, likely associated with solar rotation and the evolution of active regions. In contrast, the D2 series exhibits a markedly simplified spectral structure, characterized by a dominant short-period peak (~2–3 days), suggesting that second-order differencing effectively suppresses longer-term variability and isolates rapid, transient dynamics.

To further account for the nonstationary nature of solar activity, continuous wavelet transforms are applied to both D1 and D2 series. The wavelet power spectrum of D1 reveals pronounced, time-dependent energy within the 2–30 day band, with intermittent intensification during solar maxima (approximately 2001–2003, 2013–2015, and 2022–2023). This indicates that short-term periodicities evolve in tandem with the solar cycle rather than remaining constant. The D2 wavelet spectrum highlights even higher-frequency variability, with power concentrated below ~15 days and exhibiting localized, fragmented structures associated with abrupt fluctuations in solar activity.

Overall, the comparative analysis demonstrates a clear scale-dependent decomposition of solar dynamics. D1 captures structured short-term variability while preserving intermediate periodicities, whereas D2 isolates high-frequency, transient fluctuations linked to rapid changes in sunspot activity. These findings underscore the inherently nonstationary and multi-scale nature of solar variability and illustrate the effectiveness of combining derivative-based transformations with spectral and wavelet techniques to disentangle the temporal complexity of sunspot behavior.